

Example 1 :

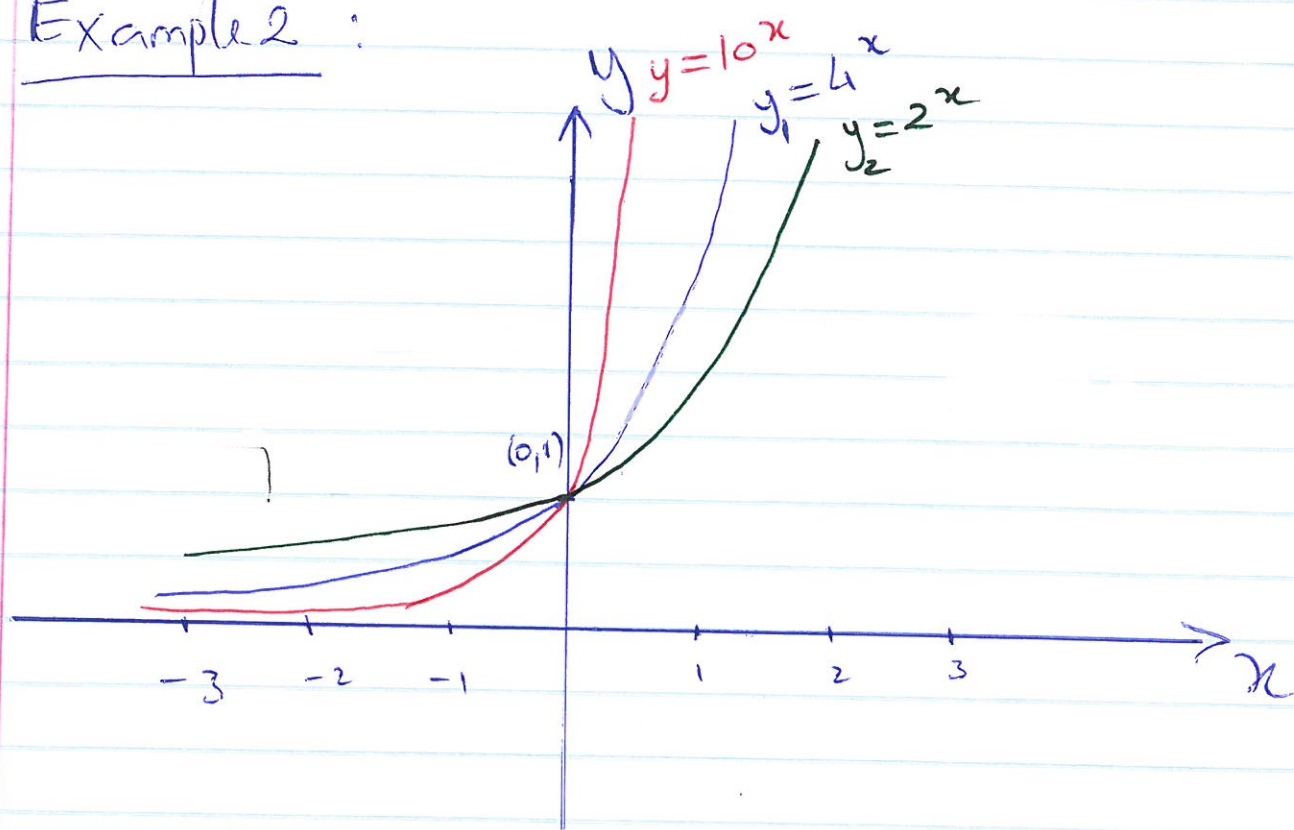
$$(a) \frac{3^{-4}}{4^{-8}} = \frac{4^8}{3^4}$$

$$(b) \frac{1}{5\sqrt{x^3}} = \frac{1}{x^{\frac{3}{5}}} = x^{-\frac{3}{5}}$$

$$(c) b^8 (2b)^3 = b^8 \times 8 \cdot b^3 = 8b^{11}$$

$$(d) \frac{\sqrt{a}\sqrt{b}}{\sqrt[4]{ab}} = \frac{(ab^{\frac{1}{2}})^{\frac{1}{2}}}{(ab)^{\frac{1}{4}}} = \frac{a^{\frac{1}{2}} b^{\frac{1}{4}}}{a^{\frac{1}{4}} b^{\frac{1}{4}}} = a^{\frac{1}{2} - \frac{1}{4}} = a^{\frac{1}{4}}$$

Example 2 :



$$y_1 = 4^x > y_2 = 2^x \text{ for all } x > 0$$

$$y_1 = 4^x < y_2 = 2^x \text{ for all } x < 0$$

Hence

If $a > b > 0$

$$a^x \geq b^x \text{ if } x \in [0, +\infty)$$

$$a^x \leq b^x \text{ if } x \in (-\infty, 0]$$

Example 1 (Continues)

$$(e) \frac{3^x \cdot (3^x)^6}{3^{4x}} = 3^x \cdot \cancel{3^{6x}} \cdot 3^{-4x} = 3^{7x-4x} = 3^{3x} = 27^x$$

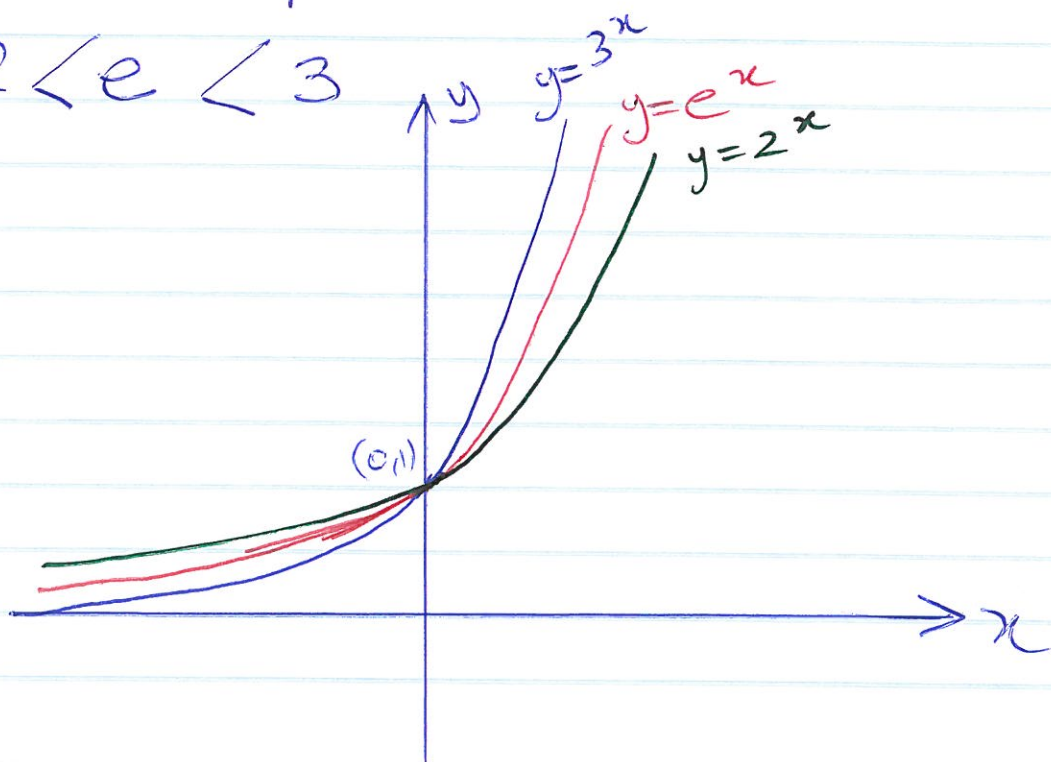
$$(f) \frac{(e^x \cdot e^y)^4}{(e^e)^5} = \frac{e^{4(x+y)}}{e^{5e}} = e^{4(x+y)-5e}$$

The number "e" in example 1 (f) is a special irrational number.

$$e \approx 2.71828$$

Therefore $f(x) = e^x$ is called the natural exponential function.

$$2 < e < 3$$



Example 3 : Using the graph of $y = e^x$

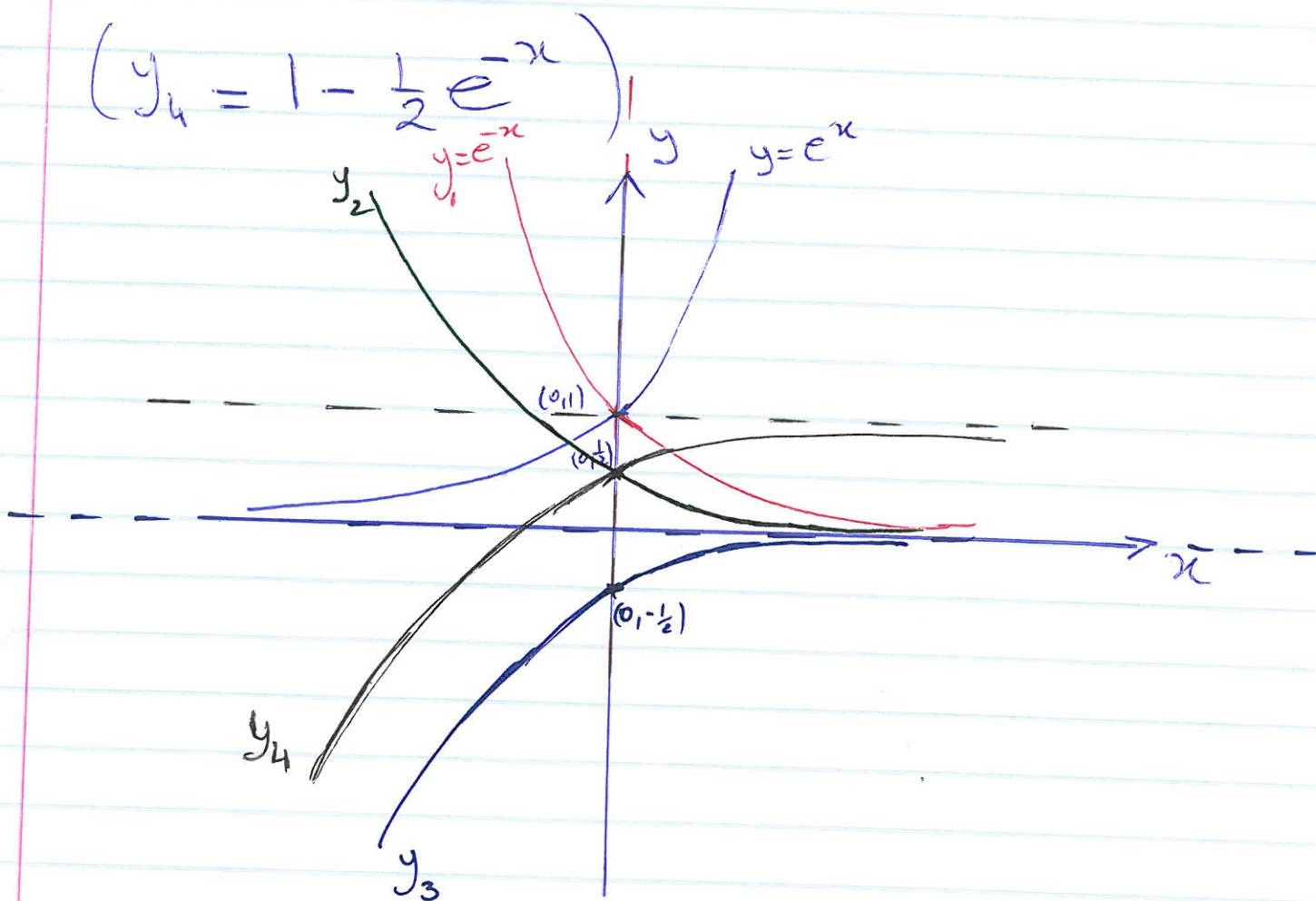
Sketch $y = 1 - \frac{1}{2}e^{-x}$.

steps:
 $y_1 = f(-x)$; $y_2 = \frac{1}{2}f(-x)$; $y_3 = -\frac{1}{2}f(-x)$

$$y_4 = 1 - \frac{1}{2}f(-x)$$

~~again~~ ~~for x < 0~~

$$(y_4 = 1 - \frac{1}{2}e^{-x})$$



Example 4 : sketch $y = e^{(x-3)}$

